

Python for AI & Machine Learning

Class 27 – Introduction to Machine Learning with Scikit-learn

Course: Python for AI & Machine Learning

Module: Machine Learning Fundamentals

Duration: 2 Hours (Theory + Practical)

Learning Objectives

By the end of this session, students will be able to:

- Understand the concept of Machine Learning.
 - Differentiate between Artificial Intelligence, Machine Learning, and Deep Learning.
 - Learn the different types of Machine Learning.
 - Understand the Machine Learning workflow.
 - Install and import the Scikit-learn library.
 - Prepare datasets for Machine Learning.
 - Train and test a simple Machine Learning model.
 - Evaluate model performance.
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Revision

In the previous class, we learned:

- Advanced Data Visualization
- Seaborn Library
- Box Plots
- Violin Plots
- Heatmaps
- Pair Plots
- Correlation Matrix
- Count Plots

Today we begin the most exciting part of the course:

Introduction to Machine Learning

What is Artificial Intelligence (AI)?

Artificial Intelligence (AI) is the ability of a computer system to perform tasks that normally require human intelligence.

Examples:

- Voice Assistants
 - Self-driving Cars
 - Chatbots
 - Face Recognition
 - Recommendation Systems
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What is Machine Learning?

Machine Learning (ML) is a branch of Artificial Intelligence that enables computers to learn from data without being explicitly programmed.

Instead of writing rules manually, we provide data, and the computer learns patterns from that data.

Real-Life Examples of Machine Learning

- Email Spam Detection
 - Netflix Movie Recommendations
 - YouTube Video Suggestions
 - Google Translate
 - Online Shopping Recommendations
 - Credit Card Fraud Detection
 - Face Unlock in Smartphones
 - Medical Disease Prediction
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AI vs Machine Learning vs Deep Learning

Artificial Intelligence	Machine Learning	Deep Learning
Broad field of creating intelligent systems	Subset of AI that learns from data	Subset of ML using neural networks
Rule-based and learning-based	Learns from datasets	Learns using deep neural networks
Larger concept	Medium scope	Specialized area
Example: Chatbots	Example: Spam Detection	Example: Image Recognition

Types of Machine Learning

Machine Learning is divided into four major categories:

1. Supervised Learning

The model learns from **labeled data**.

Example:

- Student Marks → Grade
- House Features → House Price

Algorithms:

- Linear Regression
 - Logistic Regression
 - Decision Tree
 - Random Forest
 - Support Vector Machine (SVM)
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2. Unsupervised Learning

The model learns from **unlabeled data**.

Applications:

- Customer Segmentation
- Market Basket Analysis
- Fraud Detection

Algorithms:

- K-Means Clustering
 - DBSCAN
 - Hierarchical Clustering
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3. Semi-Supervised Learning

Uses a combination of labeled and unlabeled data.

Applications:

- Medical Image Analysis
- Speech Recognition
- Document Classification

4. Reinforcement Learning

The model learns by interacting with an environment and receiving rewards or penalties.

Applications:

- Robotics
 - Self-driving Cars
 - Game Playing (Chess, Go)
 - Autonomous Drones
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Machine Learning Workflow

A typical Machine Learning project follows these steps:

1. Collect Data
 2. Clean Data
 3. Explore Data
 4. Select Features
 5. Split Dataset
 6. Train Model
 7. Test Model
 8. Evaluate Performance
 9. Deploy Model
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Installing Scikit-learn

Scikit-learn is one of the most widely used Machine Learning libraries in Python.

Install it using:

```
pip install scikit-learn
```

Verify installation:

```
pip show scikit-learn
```

Importing Libraries

```
import pandas as pd
import numpy as np

from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
```

Understanding Features and Labels

Suppose we want to predict house prices.

Size	Bedrooms	Price
1000	2	40,00,000
1500	3	55,00,000
1800	3	65,00,000

Features (Input)

- Size
- Bedrooms

Represented as:

X

Label (Output)

- Price

Represented as:

y

Sample Dataset

```
import pandas as pd

data = {
```

```
"Hours": [1,2,3,4,5],  
"Marks": [40,50,60,75,90]  
}  
  
df = pd.DataFrame(data)  
  
print(df)
```

Separating Features and Labels

```
X = df[["Hours"]]  
  
y = df["Marks"]
```

Splitting the Dataset

Training data is used to teach the model.

Testing data is used to evaluate the model.

```
from sklearn.model_selection import train_test_split  
  
X_train, X_test, y_train, y_test = train_test_split(  
  
    X,  
    y,  
    test_size=0.2,  
    random_state=42  
  
)
```

Training a Machine Learning Model

```
from sklearn.linear_model import LinearRegression  
  
model = LinearRegression()  
  
model.fit(X_train, y_train)
```

The `fit()` method trains the model.

Making Predictions

```
prediction = model.predict(X_test)

print(prediction)
```

Predicting New Values

```
new_data = [[6]]

result = model.predict(new_data)

print(result)
```

The model predicts the marks for a student who studied for 6 hours.

Evaluating the Model

```
from sklearn.metrics import mean_absolute_error

prediction = model.predict(X_test)

error = mean_absolute_error(y_test, prediction)

print(error)
```

Lower error indicates better model performance.

Understanding Accuracy

Machine Learning models are evaluated using different metrics.

For Regression:

- Mean Absolute Error (MAE)
- Mean Squared Error (MSE)
- Root Mean Squared Error (RMSE)
- R^2 Score

For Classification:

- Accuracy
 - Precision
 - Recall
 - F1 Score
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AI Example 1 – Student Marks Prediction

```
import pandas as pd
from sklearn.linear_model import LinearRegression

data = {
    "Hours": [1,2,3,4,5],
    "Marks": [40,50,60,75,90]
}

df = pd.DataFrame(data)

X = df[["Hours"]]
y = df[["Marks"]]

model = LinearRegression()

model.fit(X,y)

print(model.predict([[6]]))
```

AI Example 2 – House Price Prediction

```
from sklearn.linear_model import LinearRegression

# Features:
# Area, Bedrooms

# Label:
# House Price
```

Linear Regression is commonly used to estimate property prices.

AI Example 3 – Employee Salary Prediction

Predict salary based on:

- Experience
- Education
- Skill Level

Algorithms:

- Linear Regression
 - Random Forest Regressor
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AI Example 4 – Sales Forecasting

Predict future sales using historical monthly sales data.

Applications:

- Inventory Management
 - Business Planning
 - Demand Forecasting
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Advantages of Scikit-learn

- Easy to learn
 - Simple API
 - Large collection of algorithms
 - Excellent documentation
 - Fast model training
 - Works seamlessly with NumPy and Pandas
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Best Practices

- Always clean data before training.
 - Split data into training and testing datasets.
 - Use appropriate evaluation metrics.
 - Avoid overfitting.
 - Train with sufficient data.
 - Validate model performance before deployment.
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Practical Exercise 1

Create a student dataset.

Predict marks based on study hours using Linear Regression.

Practical Exercise 2

Split a dataset into:

- Training Data
- Testing Data

Display the size of each dataset.

Practical Exercise 3

Train a Linear Regression model.

Predict marks for students studying:

- 4 Hours
 - 6 Hours
 - 8 Hours
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Practical Exercise 4

Calculate the Mean Absolute Error (MAE) of your model.

Practical Exercise 5

Create a small dataset for house prices.

Train a model to predict the price based on house size.

Assignment

Student Performance Prediction System

Develop a Machine Learning application that:

- Reads a CSV file containing student study hours and marks.
- Splits the dataset into training and testing sets.
- Trains a Linear Regression model.
- Predicts marks for new students.
- Calculates the Mean Absolute Error (MAE).
- Displays actual vs predicted marks.

Bonus Challenge:

Visualize the regression line using Matplotlib.

Interview Questions

1. What is Machine Learning?
 2. How is Machine Learning different from Artificial Intelligence?
 3. What are the four types of Machine Learning?
 4. What is Scikit-learn?
 5. What are Features and Labels?
 6. Why do we split a dataset?
 7. What is the purpose of the `fit()` method?
 8. What does the `predict()` method do?
 9. What is Mean Absolute Error (MAE)?
 10. Name five real-world applications of Machine Learning.
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Summary

Today we learned:

- Introduction to Machine Learning
- AI vs ML vs Deep Learning
- Types of Machine Learning
- Machine Learning Workflow
- Installing Scikit-learn
- Features and Labels
- Training and Testing Data
- Linear Regression
- Model Training
- Predictions
- Model Evaluation
- AI Applications

This class marks the beginning of practical Machine Learning. Understanding these fundamentals is essential before moving on to advanced algorithms such as Decision Trees, Random Forests, Support Vector Machines, and Neural Networks.

Next Class Preview

Class 28 – Linear Regression in Depth

Topics to be covered:

- Understanding Linear Regression
- Regression Equation
- Slope and Intercept
- Simple vs Multiple Linear Regression
- Training a Regression Model
- Visualizing the Regression Line
- Model Evaluation (MAE, MSE, RMSE, R^2 Score)
- Real-world Case Studies
- Practical Machine Learning Projects